

4 Fresh pineapple contains an enzyme called bromelain.

A student thought that fresh pineapple juice might affect the growth rate of bacteria.

The student formed the following hypothesis:

E. coli growing in liquid culture will have a slower growth rate if fresh pineapple juice is added to the liquid culture.

Bacteria can be grown in liquid culture using nutrient broth as the growth medium.

Plan an investigation to find evidence to support or reject this hypothesis.

- (a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(3)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) Devise a detailed method, including how you would control and monitor important variables.

(8)



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

(Total for Question 4 = 17 marks)

TOTAL FOR PAPER = 50 MARKS